Inversion6
A DIVISION OF TRG

Microsoft Quick Assist Forensics

Queen City Con 2025

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<QUEEN CITY CONFERENCE>

V1.2

Tyler Hudak

Director of Incident Response

- 25+ years in Information Security and Incident Response
- Extensive experience in leading and responding to a variety of incidents
- Experience in diverse verticals including healthcare, manufacturing, technology, and financial services
- Accomplished speaker and trainer at local, national, and online conferences

Agenda

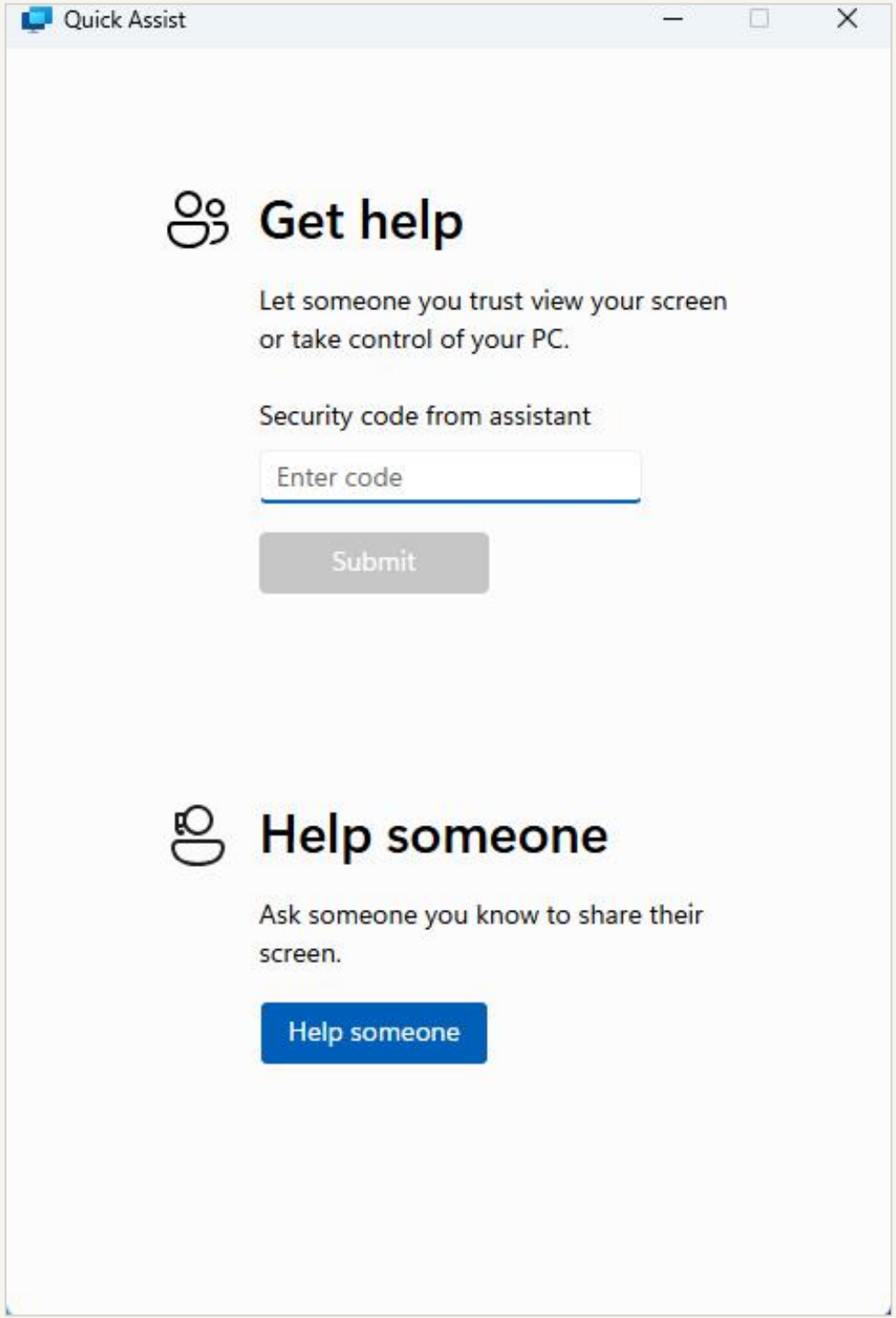
What is MS Quick Assist?

Why do we care?

How does it work?

Forensic Artifacts

Forensic Investigations



What are RMMs?

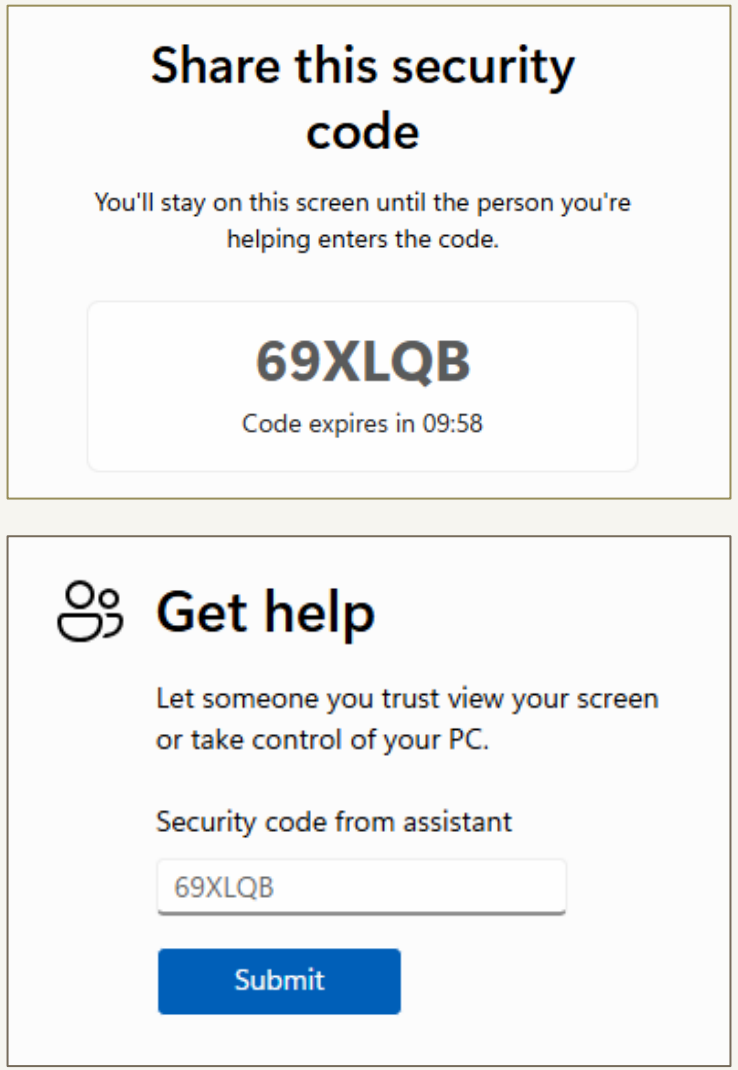
- Remote monitoring and management (RMM) tool
- Give system access to IT to troubleshoot issues

Can you connect to my computer to fix it?

Sure!

Microsoft Quick Assist

Sharer



Helper

Why are we talking about MS Quick Assist?

Microsoft warns of hacker misusing Quick Assist in Black Basta ransomware attacks

Threat researchers say a financially-motivated attacker has deployed the tool in social-engineering attacks since April.

NEWS 3 MAR 2025

Attackers Leverage Microsoft Teams and Quick Assist for Access

Research Threat intelligence Microsoft Defender Social engineering / phishing · 10 min read

Threat actors misusing Quick Assist in social engineering attacks leading to ransomware

By Microsoft Threat Intelligence

How a new malware attack turns Microsoft Teams against you

A new malware campaign uses live Microsoft Teams calls where attackers impersonate IT helpdesk staff to trick employees

by Emre Çıtak — July 18, 2025 in Research

Quick Assist Social Engineering Attacks

Attacker

Victim subjected to mail bomb attack, Teams messages, etc.

Attacker poses as “helpdesk” and calls, emails, or Teams victim

Victim convinced to connect to Attacker with Quick Assist

Attacker runs code to establish foothold and persistence

Victim

How does Quick Assist work?

From a Technical Point of View

Quick Assist Limitations

Sharer must consent to letting *helper* control system

Helper can:

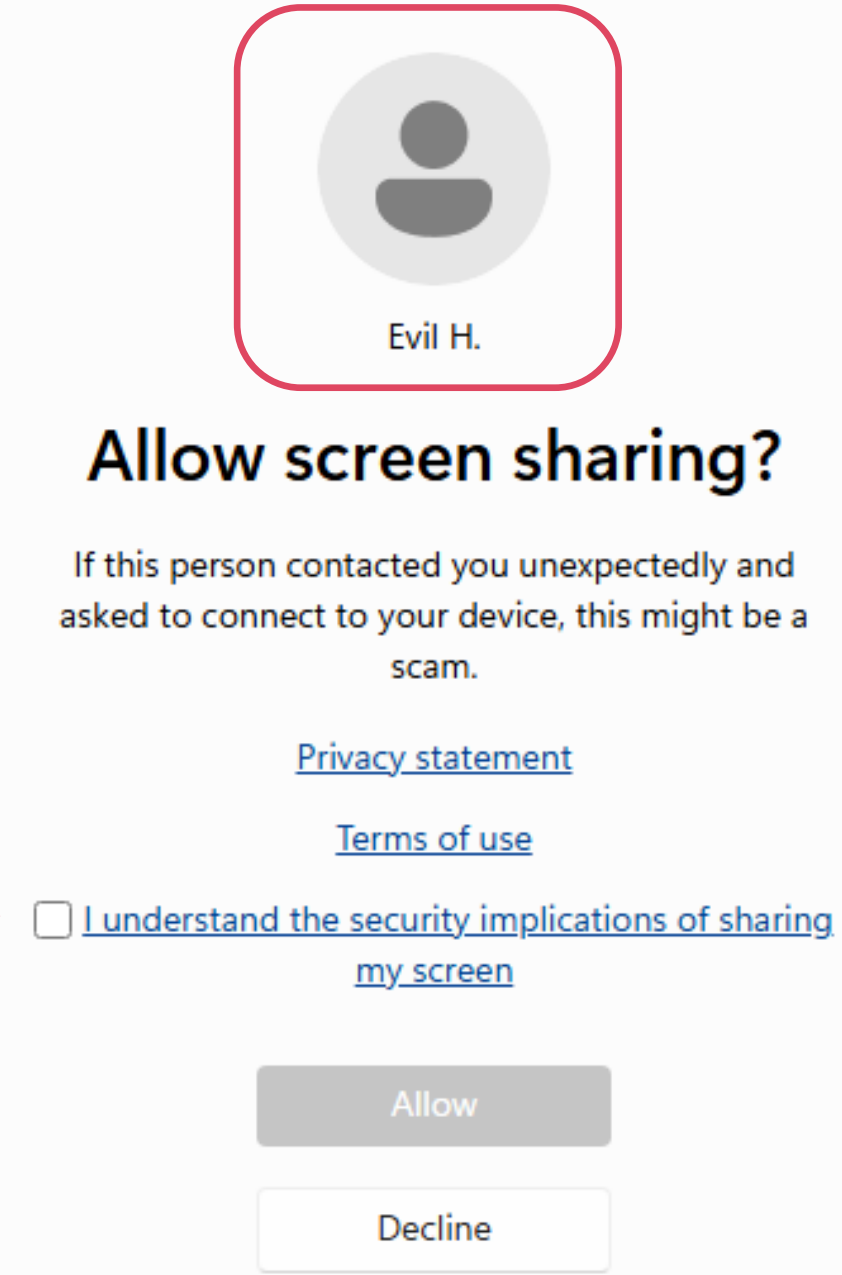
- Control system (with consent)
- Chat

Helper cannot:

- Transfer files natively
- Copy/paste through QA
- Click through UAC prompts

Consent

Helper must allow Sharer access



Two Versions

C:\windows\system32\quickassist.exe

- Location when installed by default
- Mostly Windows 10 systems

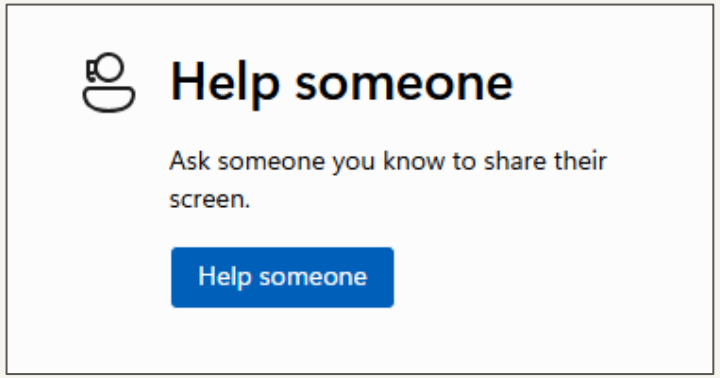
C:\Program Files\WindowsApps\MicrosoftCorporationII.QuickAssist_[VERSION]__ 8wekyb3d8bbwe\Microsoft.RemoteAssistance.QuickAssist\QuickAssist.exe

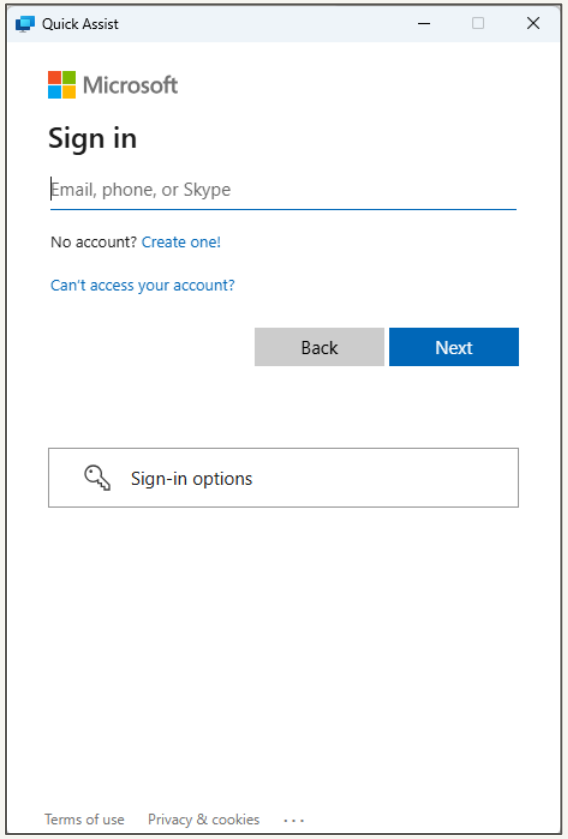
- [VERSION] is QA version #
- Windows 11 default installs and installed from Microsoft Store

QA is NOT the same as Microsoft InTune Remote Help

Authentication

- Sharer is logged into their computer already
- Helper authenticates to Microsoft Account or Entra ID





QuickAssist Execution Flow

QuickAssist.exe	< 0.01	13,228 K	49,304 K	16892 Quick Assist Component	Microsoft Corporation	TRUWEST
msedgewebview2.exe		39,916 K	127,332 K	31328 Microsoft Edge WebView2	Microsoft Corporation	TRUWEST
msedgewebview2.exe		6,372 K	18,000 K	29580 Microsoft Edge WebView2	Microsoft Corporation	TRUWEST
msedgewebview2.exe						
msedgewebview2.exe						
msedgewebview2.exe						
msedgewebview2.exe						
msedgewebview2.exe						
Notepad.exe						
slack.exe						
slack.exe						
slack.exe						
slack.exe		23,836 K	72,964 K	18296 Slack	Slack Technologies Inc.	TRUWEST

Command Line:
"C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgewebview2.exe" --embedded-browser-webview=1 --webview-exe-name=QuickAssist.exe --webview-exe-version=10.3.10076.1000 --user-data-dir="C:\Users\THudak\AppData\Local\Temp\RemoteHelp\EBWebView" --noerrdialogs --embedded-browser-webview-dpi-awareness=2 --accept-lang=en-US --disable-features=msSmartScreenProtection --enable-features=msSingleSignOnOSForPrimaryAccountIsShared --mojo-named-platform-channel-pipe=16892.16184.3580979382755180835 /pfhostedapp:33e8bb400e50c73cae1eeadc0b7087725a40d137
Path:
C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgewebview2.exe
Package:
MicrosoftCorporationII.QuickAssist_2.0.29.0_x64__8wekyb3d8bbwe

quickassist.exe

msedgewebview2.exe --webview-exe-name=quickassist.exe

Many msedgewebview2.exe
(may not have webview-exe-name)

Quick Assist Session Program Execution

- Programs executed in QA session are done under normal Windows parent-child
- NOT under quickassist.exe or msedgewebview2.exe

Executed in QA session

Parent is Explorer.exe

Process	Description	Life Time
Explorer.EXE (4152)	Windows Explorer	
SecurityHealthSystray.exe (8556)	Windows Security notification icon	
vmtoolsd.exe (8532)	VMware Tools Core Service	
msedge.exe (1788)	Microsoft Edge	
Regshot-x64-Unicode-dbg.exe (9412)	Regshot 64-Bit Unicode	
Procmon64.exe (3092)	Process Monitor	
powershell.exe (7984)	Windows PowerShell	
conhost.exe (9144)	Console Window Host	
QuickAssist.exe (7156)	Quick Assist Component	
QuickAssist.exe (3080)	Quick Assist Component	
msedgewebview2.exe (1472)	Microsoft Edge WebView2	
Notepad.exe (5196)		
cmd.exe (6720)	Windows Command Processor	
Conhost.exe (2000)	Console Window Host	
OneDrive.exe (6220)	Microsoft OneDrive	

Network Communications

- Remote Desktop Protocol (RDP) is the underlying protocol
- Communicates over TCP/443 (TLS 1.2) not TCP/3389

Sharer

https://remoteassistance.support.services.microsoft.com

Helper

URL	Description
.microsoft.com	Accessible Rich Internet Applications (ARIA) service for providing accessible experiences to users.
*.cc.skype.com	Required for Azure Communication Service.
*.events.data.microsoft.com	Required diagnostic data for client and services used by Quick Assist.
*.flightproxy.skype.com	Required for Azure Communication Service.
*.live.com	Required for logging in to the application (MSA).
*.monitor.azure.com	Required for telemetry and remote service initialization.
*.registrar.skype.com	Required for Azure Communication Service.
*.support.services.microsoft.com	Primary endpoint used for Quick Assist application
*.trouter.skype.com	Used for Azure Communication Service for chat and connection between parties.
aadcdn.msauth.net	Required for logging in to the application (Microsoft Entra ID).
edge.skype.com	Used for Azure Communication Service for chat and connection between parties.
login.microsoftonline.com	Required for Microsoft sign-in service.
remoteassistanceprodacs.communication.azure.com	Used for Azure Communication Service for chat and connection between parties.
turn.azure.com	Required for Azure Communication Service.

Quick Assist Investigations

Forensic Investigation Questions…

Is this the Helper or Sharer’s PC?

When did QA session start? End?

Was consent given?

User / IP Address of Helper

What did Helper execute/access/etc.?

Event Logs

System32 Version

- Nothing

Win 11 / Microsoft Store Version

- Logged in Application.evtx
- Source: “Quick Assist”
- Event ID 0
- Mostly truncated WebView JSON debug messages

Application Event Logs

Event Source

Event ID

QA Start of Execution

Event 0, Quick Assist

General | Details

Friendly View | XML View


```
- <Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event">
  - <System>
      <Provider Name="Quick Assist" />
      <EventID Qualifiers="0">0</EventID>
      <Version>0</Version>
      <Level>4</Level>
      <Task>0</Task>
      <Opcode>0</Opcode>
      <Keywords>0x80000000000000</Keywords>
      <TimeCreated SystemTime="2025-01-07T20:47:03.6282808Z" />
      <EventRecordID>1250</EventRecordID>
      <Correlation />
      <Execution ProcessID="3080" ThreadID="0" />
      <Channel>Application</Channel>
      <Computer>win11</Computer>
      <Security />
    </System>
  - <EventData>
      <Data>QuickAssist.exe launched</Data>
    </EventData>
  </Event>
```

WebView Debug JSON Event Logs - Request

Command **Context and Response of Command**

Info: {"command":"forwardtoagent", "context":{"command":"requestresponse", "context":{"responsename":"beginsharing", "success":true,
F26F996F3B2","connectionstring":"<E><A KH=\"HsuCRhNo+StfLgFpgkMdl8PimWc=\" KH2=\"sha256:Wytkz+9NCaeQ3SOhnBOoRwhwK
\"MIICzjCCAbagAwIBAgIQb3hy0p1zLKlJn1O4X+SEDzANBgkqhkiG9w0BAQsFADAQ
MQ4wDAYDVQQDEwV3aW4xMTAeFw0yNTAxMD
BXdpbjExMIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKC
AQEArn6xIBfyV89hltdibr9GNA0TASYSCkvZSdmZa3qxFaFrw68LYQP6Oo
rO9sZ0fUW6zi6FjwW
tZrg8vtENlQxubV+Yz5/BeiuEAUaoKSV2PKe4E6SHoXDnvtFYqzCsTnAzLghUaNS
VsVPevhZJEaITN95nr2K+
+Jqt8swwi+aKeG
xSwFSFEymbomhbPlTsyVVt/GbHtEFNYSJ34SUoUSdgz4k8PuIkcnSGu8kokCI5xY
JcSBJWg+3QthcW8+

- Context Command "*requestresponse*" is request for action
- "*context:responsename*" contains info on the request/response or success
- Not all events are logged or may be <u>truncated</u>

WebView Debug JSON Event Logs - Response

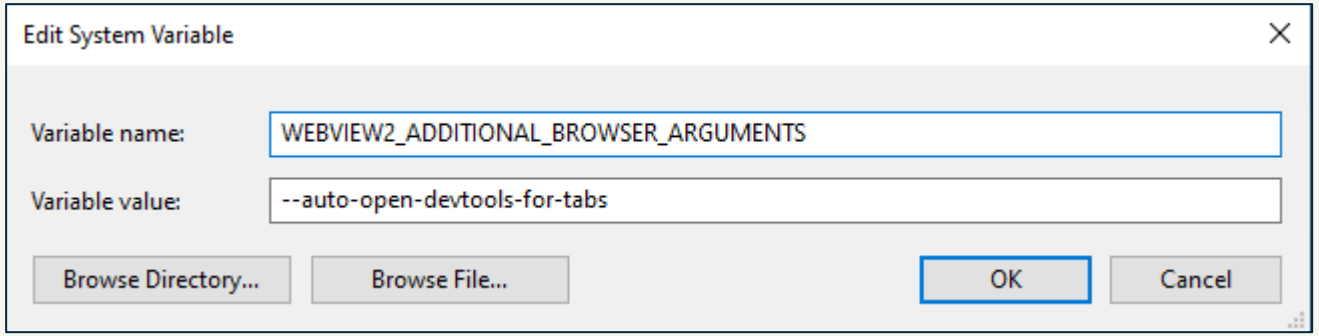
Original Request Command

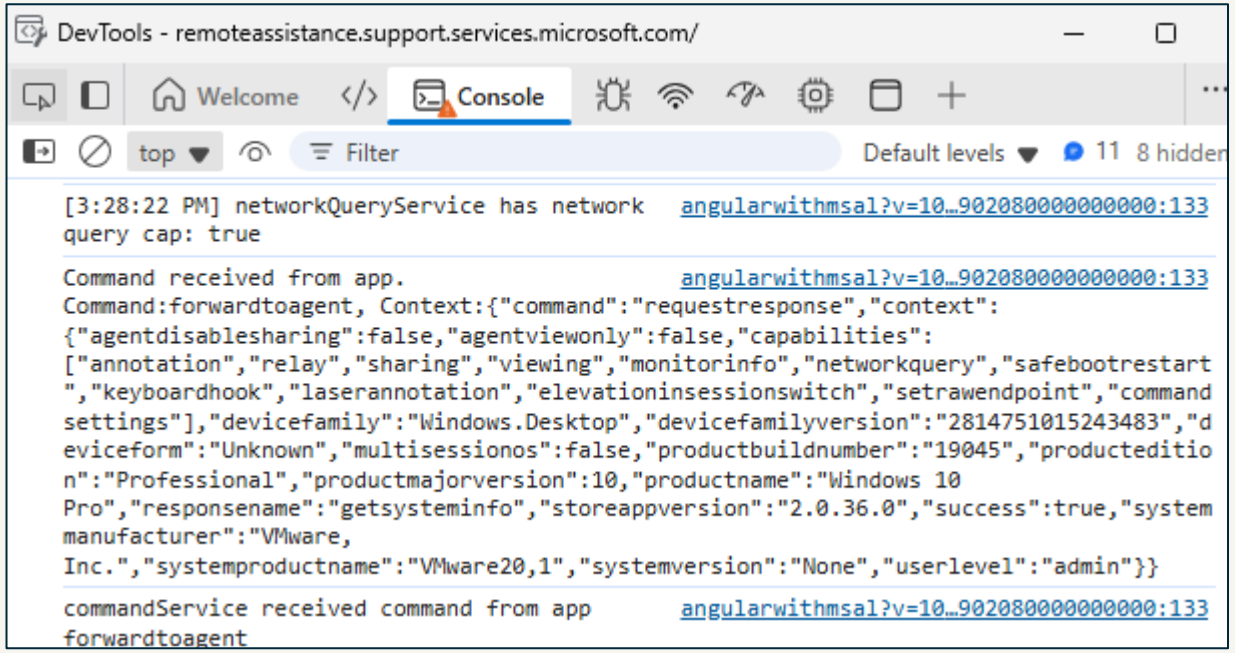
Success and Additional Data

Result: {"responsename":"beginsharing","success":true,"sessionpassword":"{CFF193D

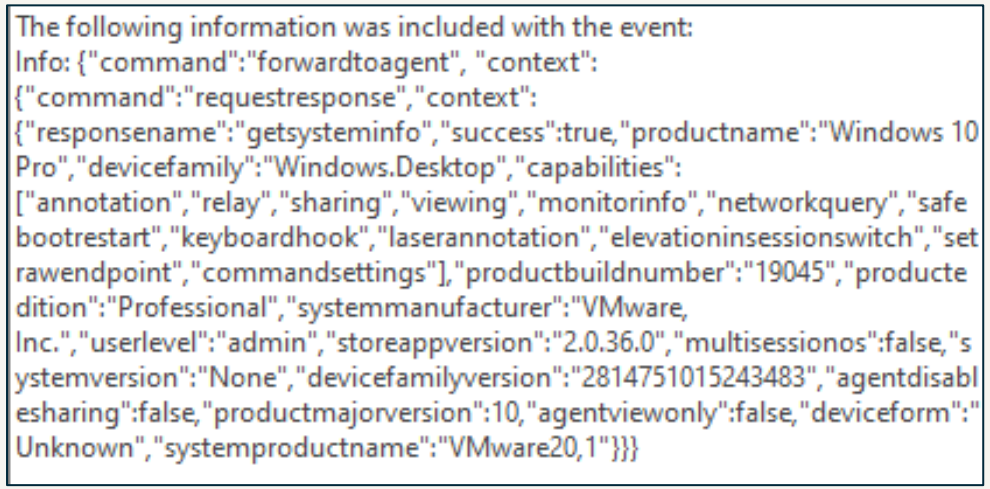
- "*responsename*" contains original request command
- Result and additional data follows

WebView Debug Messages





DevTools Console



Event Logs

Quick Assist Event Log Messages

Event	Request Context Response Name	Context Info
Quick Assist Launched	*QuickAssist.exe Launched*	
Sharer shared screen with Helper	beginsharing	
Sharer gives Consent to Helper	setsharingmode	SharingMode: FullControl
Control is cancelled or removed	setsharingmode	SharingMode: View
Sharing stops	endsharing	
Quick Assist closed	sendappclose	

%APPDATA%\Local\Temp\RemoteHelp\EBWebView (Sharer)

- MS Edge Cache Files from msedgewebview2.exe
- Analyze with Hindsight (https://github.com/obsidianforensics/hindsight)
- *https://remoteassistance.support.services.microsoft.com*
  - **First instance occurs shortly after QA launched**
- *https://remoteassistance.support.services.microsoft.com/screenshare*
  - **Shortly after screen sharing enabled or sharing code is entered**
- *https://remoteassistance.support.services.microsoft.com/status/ended*
  - **Sharing connection has been closed**

Type	Timestamp (UTC)	URL	Title / Name / Status
preference (session)	2025-01-07 20:47:05.095		Session event log [in Prefere
site setting (dips)	2025-01-07 20:47:07.348	microsoft.com	first_site_storage_time
url	2025-01-07 20:47:07.399	https://remoteassistance.support.services.microsoft.com/	Microsoft Quick Assist
site setting (modified)	2025-01-07 20:47:07.400	https://[*.]microsoft.com,*	cookie_controls_metadata [i
url	2025-01-07 20:47:24.148	https://remoteassistance.support.services.microsoft.com/screenshare	Microsoft Quick Assist

Registry Artifacts - Sharer

Key	Value	Data	Notes
HKCU\Software\Classes\Local Settings\Software\Microsoft\ Windows\CurrentVersion\ AppModel\SystemAppData\ MicrosoftCorporationII.QuickAssist_8wekyb3d8b bwe	WasEverActivated	1	Indicates use of QuickAssist
…\HAM\AUI\App\V1\LU\	PCT PTT	(hex)	Unknown purpose

Examine Registry key last modified time for last QA execution

Username of Helper


```
[2:00:55 PM] Channels command received. Command:        angularwithmsal?v=10…9789760000000000:133
{"command":"beginsharing","context":{"connectionid":"{20f09b5a-f0ae-4f50-85ad-
38c5158da15e}","hostname":"rdprelayv3eastusprod-
4.relay.support.services.microsoft.com","port":443,"promptuser":true,"sharingmode":"View","corre
lationid":"{3e035140-ea20-493e-c8ef-6aa37fbfb3c0}","connecttimeout":120000,"userInfo":"
{\"Email\":\"Evil Hacker\",\"FirstName\":\"Evil\",\"LastName\":\"Hacker\",
\"IsMicrosoft\":false,\"IsMicrosoftSupport\":false,\"CompanyName\":\"\"}","responsename":
```


Username of helper is recorded in Webview Debug Messages

But truncated in event logs!


```
The following information was included with the event:
Incoming cmd Message: {"command":"beginsharing","context":{"connectionid":"{20f09b5a-f0ae-
4f50-85ad-38c5158da15e}","hostname":"rdprelayv3eastusprod-4.relay.support.services
```

QA Execution Artifacts

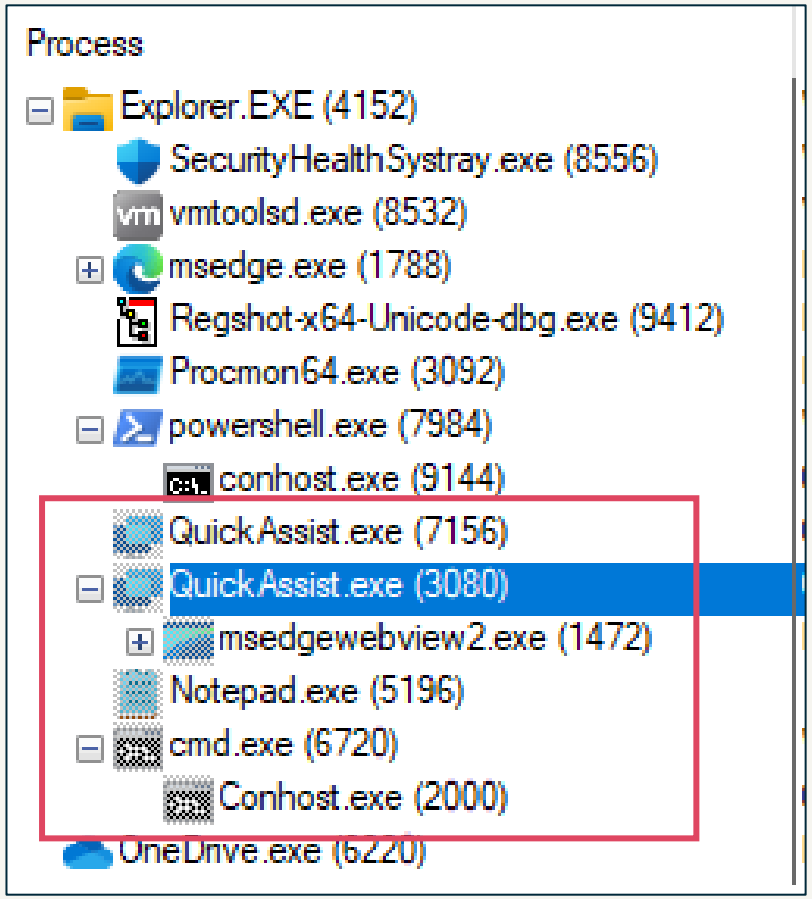
Programs executed run under normal Windows parent/child relationships

↓

All executed processes will be under the Sharer's ID

↓

Process execution logs will <u>NOT</u> show executions by the helper



Determining Attacker Actions

1. Find start of QA session

- Event logs – Sec 4688, Sysmon 1 for QuickAssist.exe
- App event log Quick Assist source “Quick Assist Launched”
- QA MS Edge history files

2. Find stop of QA session

- Application event log Quick Assist source
  - endsharing or sendappclose context
- QA MS Edge history files
- Last mod time of file system artifacts

3. Obtain “normal” evidence of execution artifacts

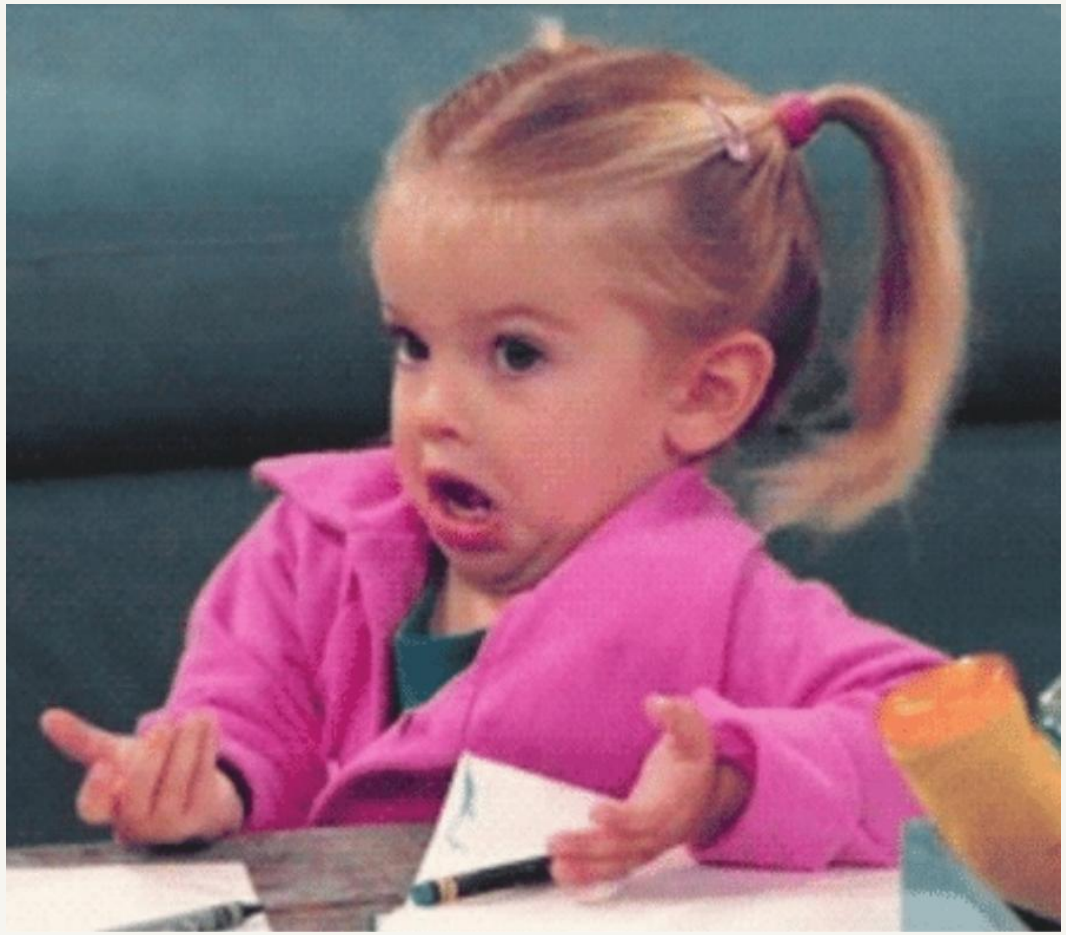
- Prefetch, SRUM, etc.
- Sec 4688 / Sysmon 1
- Registry: AmCache, UserAssist, BAM, MUICache, etc.

4. What falls in between the start and stop times?

Network Artifacts

- Traffic:
  - Goes through Microsoft servers
  - Encrypted
- Use your network telemetry
- Microsoft may data for a limited time

> ⓘ **Note**
>
> No logs are created on either the helper's or sharer's device. Microsoft can't access a session or view any actions or keystrokes that occur in the session.
>
> The sharer sees only an abbreviated version of the helper's name (first name, last initial) and no other information about them. Microsoft doesn't store any data about either the sharer or the helper for longer than three days.



Forensic Investigation Questions…

- Is this the Helper or Sharer’s PC? ✓
- When did QA session start? End? ✓
- Was consent given? ✓
- User ID / IP Address of Helper ✗
- What did Helper execute/access/etc.? ?

Prevention

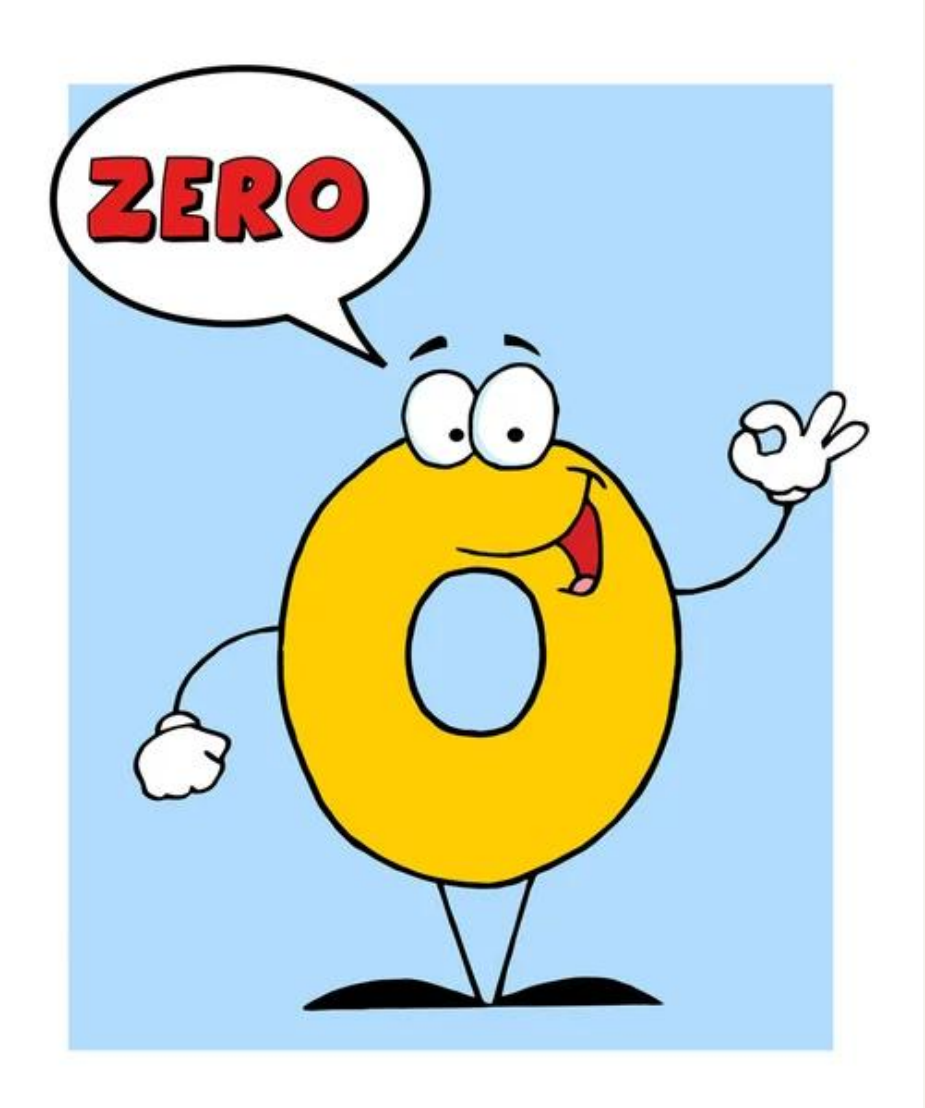
Preventing Quick Assist Social Engineering Attacks

- User awareness
  - Social engineering methods
  - Escalation procedures
  - How IT will act
- Block external users from communicating in Teams
  - https://learn.microsoft.com/en-us/microsoftteams/trusted-organizations-external-meetings-chat
- Policies on approved remote access software

Quick Assist Access Controls

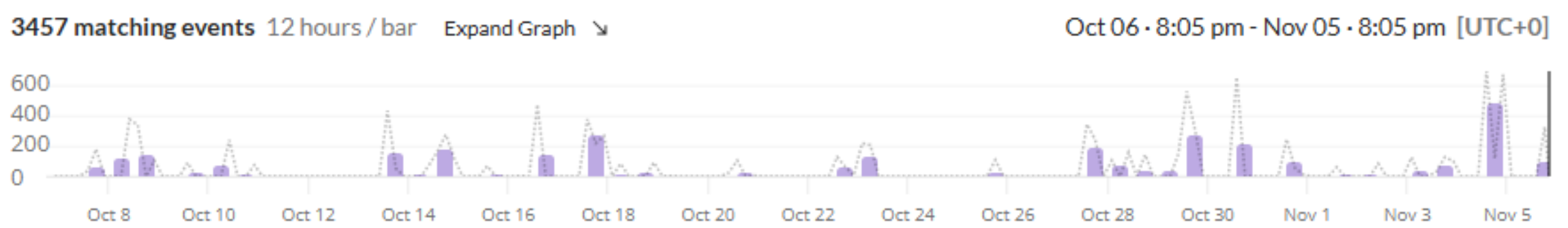
No native access controls

Unless you buy MS Intune Remote Help



Preventing Quick Assist Access

- Block access to https://remoteassistance.support.services.microsoft.com
  - **Also blocks MS Intune Remote Help**
- Uninstall and/or delete Quick Assist
  - **Need it? Upgrade to MS Store version AND delete System32 version**
- Denylist Quick Assist
- Monitor for Quick Assist execution



Resources

- Investigating Microsoft Quick Assist:
  - **https://hackuponthegale.github.io/blog/dfir/QuickAssist1**
- LOLRMM
  - **https://lolrmm.io/tools/quick_assist**
- Strontic xCyclopedia
  - **https://strontic.github.io/xcyclopedia/library/quickassist.exe-39AB5ED601B0C39DCE3B7D269847C944.html**
- Microsoft Quick Assist
  - **https://learn.microsoft.com/en-us/windows/client-management/client-tools/quick-assist**

Thank You

QUESTIONS?

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